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<https://dasma.pcu.edu.ph>

Configuring SSH Remote Access on Ubuntu Server

Objective

Set up an Ubuntu Server with SSH (Secure Shell) remote access and connect to it from a host desktop machine using Bridged Network mode.

Operating System Installation

Install Ubuntu Server

1. Start the VM and boot from the Ubuntu Server ISO
2. Follow the installation wizard:
 - Profile setup: Create a user account (e.g., server / server123)
 - Install OpenSSH server: **Check this option** (Important!)
 - Selected software: **None** (Standard Ubuntu Server)
3. Wait for installation to complete and reboot

Network Configuration

Configure Network Mode

For VirtualBox:

1. Power off the VM (if running)
2. Go to **Settings** → **Network**
3. **Adapter 1**: Enable Network Adapter
4. **Attached to**: Select **Bridged Adapter**
5. **Name**: Select your host's physical network adapter (Wi-Fi/Ethernet)
6. Click **OK**

For VMware:

1. Power off the VM
2. Go to **VM Settings** → **Network Adapter**

3. Select **Bridged** mode

4. Click **OK**

Install/Verify SSH Server

Server: Install SSH on the server (if not already installed):

- `sudo systemctl status ssh` – for checking the SSH status.
- `sudo apt update` – updating the your server.
- `sudo apt install openssh-server -y` –installing the SSH.
- `sudo systemctl enable ssh` – enabling the SSH to your unit. (SERVER/DESKTOP)
- `sudo systemctl start ssh` – starting the SSH to your unit. (SERVER/DESKTOP)

Bridged makes the server behave like a separate PC in your LAN.

1. Find the Server's IP

Inside the VM, run:

```
Sample@UBUNTU-DESKTOP:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:9b:8c:ca brd ff:ff:ff:ff:ff:ff
    inet 192.168.1.4/24 brd 192.168.1.255 scope global dynamic noprefixroute enp0s3
        valid_lft 83602sec preferred_lft 83602sec
    inet6 2001:4451:b00:2000:d76c:1af2:a10c:f36d/64 scope global temporary dynamic
        valid_lft 179068sec preferred_lft 83386sec
    inet6 2001:4451:b00:2000:a00:27ff:fe9b:8cca/64 scope global dynamic mngtmpadr proto kernel_r
        valid_lft 179068sec preferred_lft 92668sec
    inet6 fe80::a00:27ff:fe9b:8cca/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
Sample@UBUNTU-DESKTOP:~$
```

From the desktop, connect to the server: `ssh username@192.168.1.120`

2. Connect from Desktop

From your host desktop:

```
Sample@UBUNTU-DESKTOP:~$ ssh Server@192.168.1.14
Server@192.168.1.14's password:
Welcome to Ubuntu 24.04.1 LTS (GNU/Linux 6.8.0-79-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Mon Sep  8 11:24:38 PM UTC 2025

System load:                0.0
Usage of /:                  5.1% of 48.91GB
Memory usage:               5%
Swap usage:                 0%
Processes:                  139
Users logged in:            1
IPv4 address for enp0s3:    192.168.1.14
IPv6 address for enp0s3:    2001:4451:b00:2000:a00:27ff:fe57:bc4a

Expanded Security Maintenance for Applications is not enabled.

136 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Last login: Mon Sep  8 22:52:07 2025 from 192.168.1.4
Server@UBUNTU-SERVER:~$
```

Submission Requirements

1. Screenshots (labeled clearly):

- VM network settings showing Bridged Adapter
- Server IP address (output of `ip a` or `hostname -I`)
- SSH service status (`sudo systemctl status ssh`)
- Successful SSH connection from host
- Verification commands executed inside VM

2. Submission Format:

- Single PDF file
- Include name, student ID, and date
- All screenshots with brief explanation
- Upload to your portfolio website.

UBUNTU SERVER SSH REMOTE ACCESS LAB

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Course/Year/Section: BSIT 3B

Date Submitted:02/09/26

Instructor:Johneros P. Puyo

1 OBJECTIVES (5 POINTS)

Check all that apply:

- ☐ **Create and configure a Virtual Machine with Ubuntu Server**
- ☐ **Configure Bridged Network Adapter for network access**
- ☐ **Determine and verify the server's IP address**
- ☐ **Install OpenSSH Server and enable the service**
- ☐ **Establish successful SSH connection from host machine**
- ☐ **Execute verification commands to confirm remote access**

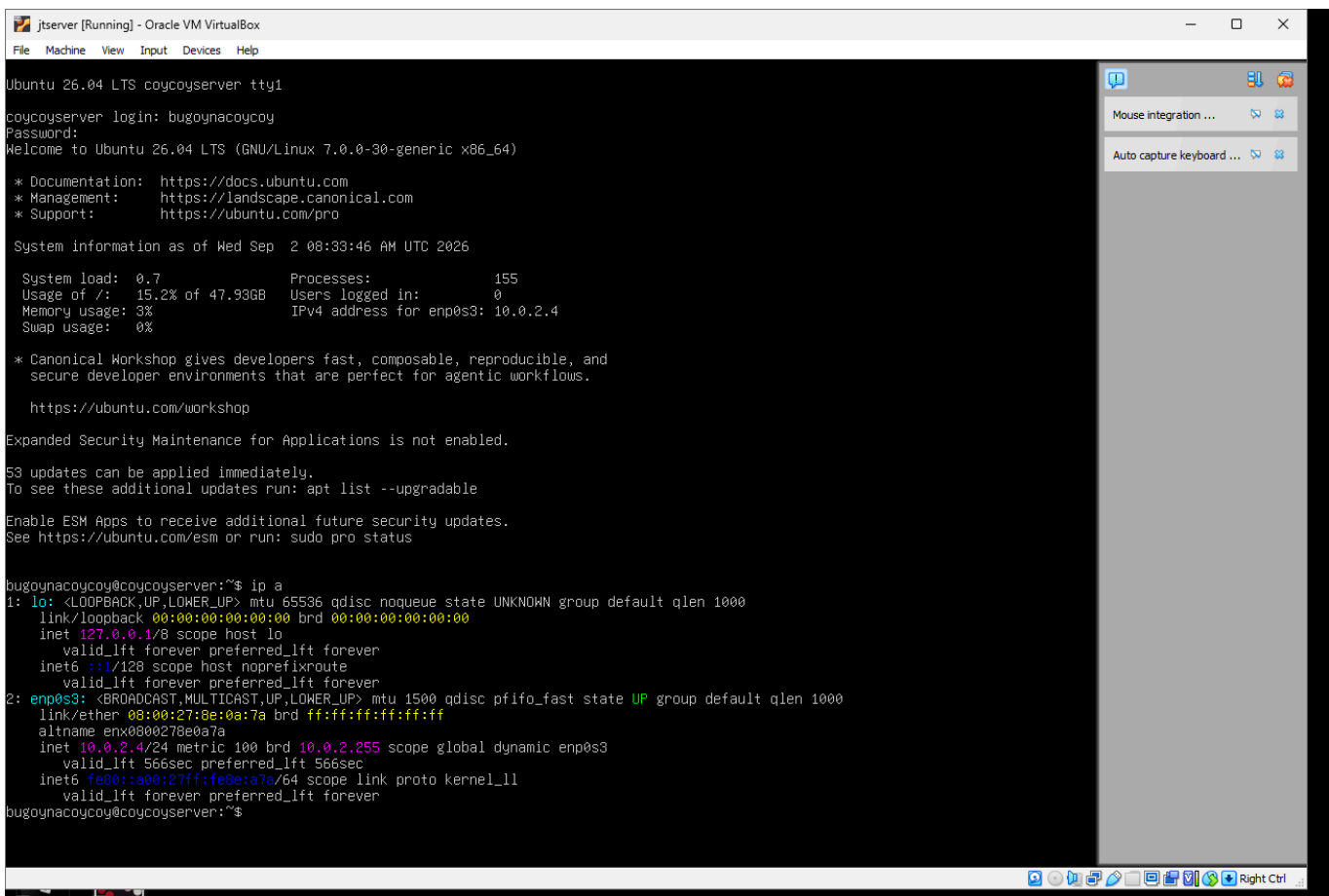
2 HARDWARE & SOFTWARE REQUIREMENTS (5 POINTS)

Item	Specification
Host Operating System	ubuntu
Virtualization Software (VirtualBox/VMware)	Oracle virtual box
Ubuntu Server Version	ubuntu -26.04 live server
VM Name	jtserver
RAM Allocated	8000
Storage Allocated	100
Network Mode	Nat network
SSH Client Used	Secure shell client

3 VM CONFIGURATION

Setting	Value
VM Name	UBUNTU-SERVER
OS Type	Ubuntu (64-bit)
Memory	8000 MB
Hard Disk	100 GB
Network Adapter	Bridged

4. NETWORK CONFIGURATION



Introduction to Cloud Comput... Cloud Computing Models - Cl... Virtualization and VM - Class... Ask Gemini

classroom.google.com/u/0/c/Od9OT10U2UMj1m/OU010TESMj4N24/details

Continue where you left off: Chrome can restore your tabs every time you restart. To turn this on, go to [Settings](#).

Classroom > Cloud Computing and Services

ubuntu os (Running) - Oracle VM VirtualBox

File Machine View Input Devices Help

Sep 2 11:56

```
bulkd@bulkd-VirtualBox:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:c5:b6:ff brd ff:ff:ff:ff:ff:ff
    inet 192.168.1.100/24 scope global enp0s3
        valid_lft forever preferred_lft forever
    inet6 fe80::27c5:b6ff:fe00:0000/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
bulkd@bulkd-VirtualBox:~$
```

Oracle VM VirtualBox Manager

House integration Auto capture key

Tools

- Ubuntu_9H Powered Off
- almsalled Powered Off
- bulkd_SAP_BUTTONING_CLIENT_DELETE Disabled
- bulkd_SAP_BUTTONING_SERVER_DELETE Disabled
- VM_DELIVERING Disabled
- sample Powered Off
- Appheap_vb Powered Off
- Tristan Disabled
- happymonkey Disabled
- almsalled Powered Off
- ubuntu os Running

General

Name: ubuntu os
Operating System: Ubuntu (64-bit)

System

Base Memory: 4096 MB
Processor: 4
Hard Disk: Optical, Floppy
Acceleration: Nested Paging, VMX Paravirtualization

Display

Video Memory: 32 MB
Graphics Controller: VBox/SGX
Accelerate Desktop Server: Disabled
Recording: Disabled

Storage

Controller: IDE
Virt. Storage Device 0: Central Drive Empty
Controller: SATA
SATA Port 0: ubuntu-os.vdi (20,00 GB)

Audio

Host Driver: Default
Controller: ICH8-AC97

Network

Adapter 1: Intel PRO/1000 MT Desktop (Bridged Adapter, Realtek PCIe GbE Family Controller)

USB

USB Controller: OHCI_EHCI
Device Filters: 0 (0 active)

Shared Folders

Name:
Description:

Preview

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Search

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